

final_practice

 This is a preview of the published version of the quiz

Started: Apr 7 at 11:11am

Quiz Instructions

I am missing a few questions from last units . Electromagnetic spectrum (order from radio waves to gamma rays) and

$$S \text{ (W/m}^2\text{)} = P \text{ (W)} / (4\pi r^2).$$

Example: if a light bulb is 60 W (total energy radiated), the amount of energy (radiation) going through an unit area at a distance r is $S = P / (4\pi r^2)$. Its the relationship between luminosity / intensity and bightness/ S .

grade on practice exam does not matter. there is a tutorial in my dropbox.



Question 1 7.7 pts

Two isolated copper plates, each of area 0.4 m^2 , carry opposite charges of magnitude 7.08×10^{-10} coulombs. They are placed opposite each other in parallel alignment, with a spacing of 2 mm . What will be the potential difference between the plates when their spacing is increased to 4 cm ?



8 V



7.6 V



3.04 V



0.4 V



3.2 V



Question 2 7.7 pts

A light bulb produces 10 W/m^2 at a distance of 5 m (so 5 m away from the source, 10J of energy flows though a square $1 \text{ m} \times 1 \text{ m}$ every second).

hint: Brightness = Luminosity / ($4\pi R^2$) . Luminosity is the power of the source. brightness is the power at a distance R per unit area.

☐

about 3000W

☐

about 30W

☐

about 300W

☐

about 30,000W

☐

Question 3 7.7 pts

A proton starting from rest travels through a potential of 1.0 kV and then moves into a uniform 0.040-T magnetic field. What is the radius of the proton's resulting orbit? ($m_{\text{proton}} = 1.67 \times 10^{-27}$ kg, $e = 1.60 \times 10^{-19}$ C)

☐

0.17 m

☐

0.19 m

☐

0.14 m

☐

0.080 m

☐

0.11 m

☐

Question 4 7.7 pts

The potential difference between the plates of an ideal air-filled parallel-plate capacitor with a plate separation of 6.0 cm is 60 V. What is the strength of the electric field between the plates of this capacitor?

☐

3600 V/m

☐

1000 V/m

☐

60 V/m

☐

2000 V/m

☐

500 V/m

☐

Question 5 7 pts

At a certain instant in time, the electric field of an electromagnetic wave in free space points in the $-z$ direction, and the magnetic field points in the $+y$ direction. In what direction is this wave traveling?
 hint: right hand rule. (V, E, B). Very Educated Beaver? (Velocity, Electric field, magnetic field)

☐

$+z$ direction

☐

$-z$ direction

☐

$+y$ direction

☐

$+x$ direction

☐

$-x$ direction

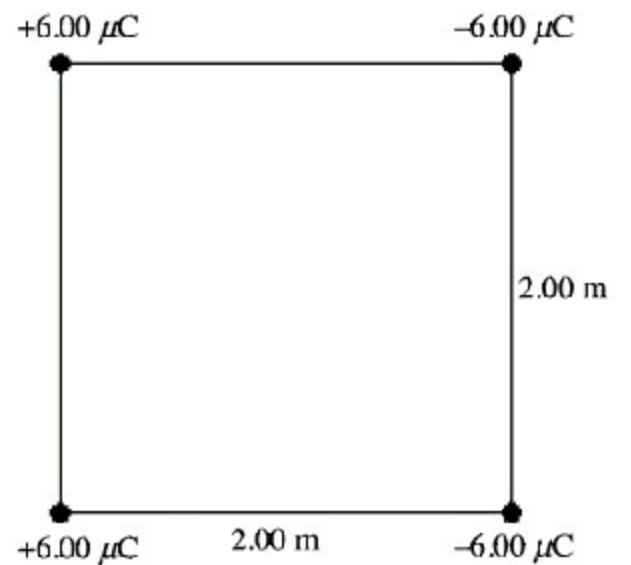
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Question 6 7.7 pts

Four point charges of magnitude $6.00 \mu\text{C}$ and of varying signs are placed at the corners of a square 2.00 m on each side, as shown in the figure. ($k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

(a) What is the electric potential (relative to infinity) at the center of this square due to these charges?

You are adding scalars. See equation for V. You have a similar question for final


☐

50 kV

☐

20 kV

☐

10 kV



0



Question 7 7.7 pts

Which one of the following lists gives the correct order of the electromagnetic waves from *longer wavelength to shorter wavelength*?



radio waves, microwaves, infrared, visible, ultraviolet, x-rays, gamma rays



radio waves, microwaves, visible, x-rays, infrared, ultraviolet, gamma rays



radio waves, infrared, x-rays, microwaves, ultraviolet, visible, gamma rays



radio waves, ultraviolet, x-rays, microwaves, infrared, visible, gamma rays



radio waves, infrared, microwaves, ultraviolet, visible, x-rays, gamma rays



Question 8 7.7 pts

Two long parallel wires carry currents of 10 A in opposite directions. They are separated by 40 cm. What is the magnitude of the magnetic field in the plane of the wires at a point that is 20 cm from one wire and 60 cm from the other? ($\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$)

hint: Its a vector sum. First draw B1 and B2. Decide if you add or subtract (right hand) then find magnitude. Remember to convert. Use right hand. the thumb is a wire. The fields will be in the screen or out of the screen



67 μT



6.7 μT



3.3 μT



1.5 μT



33 μT



Question 9 7.7 pts

If the flux of a magnetic field through a loop is given by the function $5t^3 + 5$, what is the induced emf at $t = 2 \text{ s}$? (hint: take the derivative and plug)



10V



50V



45V



60V



Question 10 7.7 pts

A $+5.0\text{-}\mu\text{C}$ point charge is placed at the 0 cm mark of a meter stick and a $-4.0\text{-}\mu\text{C}$ charge is placed at the 50 cm mark. What is the net electric field at the 30 cm mark? ($k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$) hint: an electric field is a vector. It is a vector sum.

 $9.0 \times 10^5 \text{ N/C}$  $4.0 \times 10^5 \text{ N/C}$  $1.4 \times 10^6 \text{ N/C}$  $5.0 \times 10^5 \text{ N/C}$ 

Question 11 7.7 pts

A non conducting sphere has a radius of 5cm. The charge density is $100\mu\text{C}/\text{m}^3$.

What is the electric field at a distance 3cm from the center? (about)

hint: it is not a conductor so you can have charges inside the sphere. Use gauss law.

Flux of E through a (imaginary) spherical surface of radius 3cm = $Q_{\text{inside}}/\epsilon_0$.

Q_{inside} = charge density x volume of imaginary sphere ($r=3\text{cm}$).

The flux = E x surface area of imaginary sphere (E is diverging away from the imaginary spherical area and is perpendicular to area)



210KN/C



50KN/C



110 KN/C



10KN/C



Question 12 9 pts

A flat rectangular coil with dimensions of $8.0 \text{ cm} \times 10 \text{ cm}$ is dropped from a zero magnetic field position into a 1.4-T magnetic field in 0.10 s . The coil has 60 turns and is perpendicular to the magnetic field. What is the average induced emf in the coil as a result of this action?

☐

0 V

☐

3.6 V

☐

2.4 V

☐

6.7 V

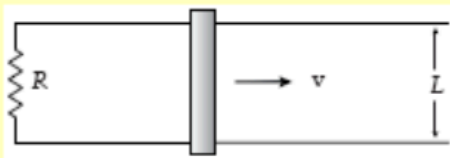
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8.6 V

⋮

Question 13 5 pts

A bar ($L = 80 \text{ cm}$) moves on two frictionless rails, as shown, in a region where the magnetic field is uniform ($B = 0.30 \text{ T}$) and into the paper. If $v = 50 \text{ cm/s}$ and $R = 60 \text{ m}\Omega$, what is the magnetic force on the moving bar?



- a. 0.48 N to the right
- b. 0.48 N to the left
- c. 0.32 N to the left
- d. 0.32 N to the right
- e. None of the above

hint: First find the direction of the induced current using lenz law. Find the emf induced and the induced current. The bar is now a current carrying wire (induced wire). So a current carrying wire inside a magnetic field is acted by a magnetic force. (ILB)

☐

0.32 N to the right

☐

0.48 N to the right

☐

0.48 N to the left

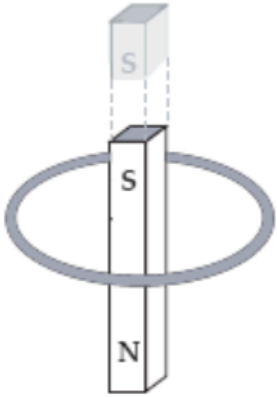
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0.32 N to the left

⋮

Question 14 1 pts

A bar magnet is dropped from above and falls through the loop of wire shown below. The north pole of the bar magnet points downward towards the page as it falls. Which statement is correct?



- The current in the loop always flows in a clockwise direction.
- The current in the loop always flows in a counterclockwise direction.
- The current in the loop flows first in a clockwise, then in a counterclockwise direction.
- The current in the loop flows first in a counterclockwise, then in a clockwise direction.
- No current flows in the loop because both ends of the magnet move through the loop.

☐

b

☐

c

☐

e

☐

d

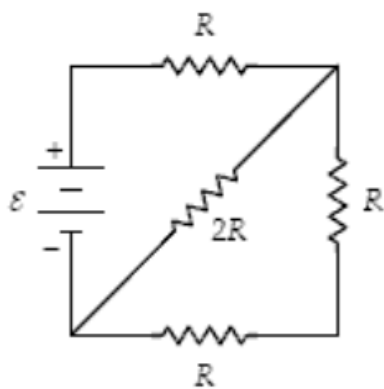
☐

a

☒

Question 15 1 pts

At what rate is thermal energy being generated in the $2R$ -resistor when $\mathcal{E} = 12 \text{ V}$ and $R = 3.0 \Omega$?



- a. 12 W
- b. 24 W
- c. 6.0 W
- d. 3.0 W
- e. 1.5 W



1.5W



6W



12W



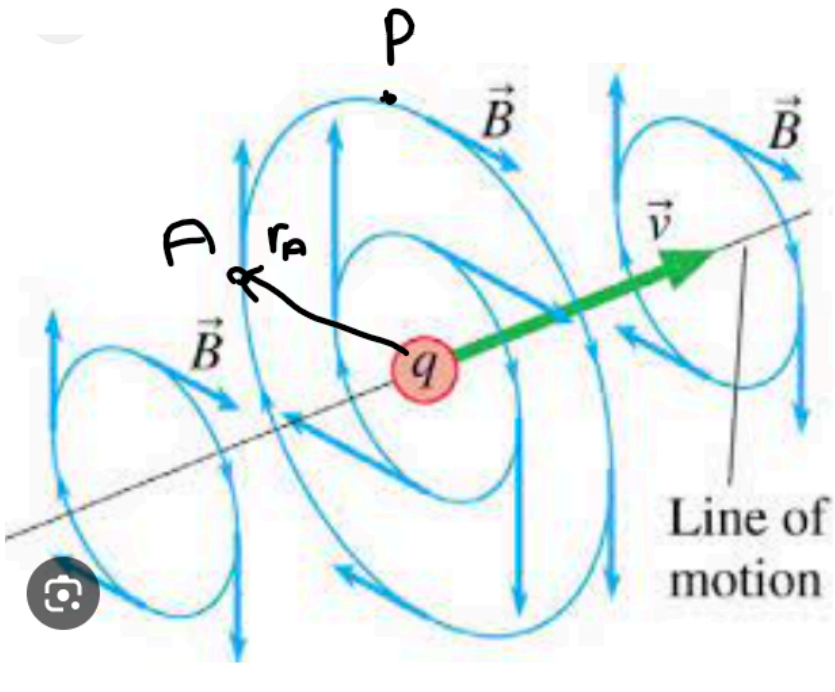
3W



24W



Question 16 1 pts



a moving charge with a velocity V is a source of magnetic field.

At point A the magnetic field is up as shown. You can use your right hand. r_A is the vector position from the charge to A. Use your 3 fingers to find direction of B. ($V \times r$ is proportional to B)

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \mathbf{r}}{|\mathbf{r}|^3}$$

☐

IDC

☐

ok

☐

What.....

☐

IDK

⋮

Question 17 1 pts

An AC generator consists of 6 turns of wire. Each turn has an area of 0.040 m^2 . The loop rotates in a uniform field ($B = 0.20 \text{ T}$) at a constant frequency of 50 Hz. What is the maximum induced emf?

- a. 13 V
- b. 2.4 V
- c. 3.0 V
- d. 15 V
- e. 4.8 V

☐

d

☐

e

☐

a

☐

b

☐

c

☒

Question 18 1 pts

A 2.0-m wire carries a current of 15 A directed along the positive x axis in a region where the magnetic field is uniform and given by $B = (30\mathbf{i} - 40\mathbf{j})$ mT. What is the resulting magnetic force on the wire?

- a. $(+1.2 \mathbf{k})$ N
- b. $(-1.2 \mathbf{k})$ N
- c. $(-1.5 \mathbf{k})$ N
- d. $(+1.5 \mathbf{k})$ N
- e. $(+0.90 \mathbf{k})$ N

The force has to be perpendicular to B and I . See right hand rule and equation for a segment in a field.

Only the component of B perpendicular to I matters. Or you can do the cross product.

☐

b

☐

c

☐

e

☐

a

☐

d

Not saved

Submit Quiz